

Learning Macroeconomic Policy via Reinforcement Learning: A Dynamic Control Approach (Demo)

Abstract

We propose a reinforcement learning (RL) framework in which a neural policy learns to conduct macroeconomic stabilization policy in a simulated dynamic economy. The economy is modeled as a stochastic, nonlinear system with lagged responses, expectations formation, and fiscal-monetary interactions. Using a Soft Actor-Critic (SAC) architecture, the agent learns continuous policy interventions that optimize a multi-objective welfare function balancing inflation, unemployment, growth, and debt sustainability. We benchmark against rule-based policies and evaluate robustness under structural shocks.

1 Introduction

Macroeconomic policy is fundamentally a dynamic control problem under uncertainty. Classical approaches (e.g., Taylor rules, DSGE optimization) rely on fixed functional forms and strong assumptions. In contrast, reinforcement learning offers a data-driven method for discovering adaptive policies in nonlinear environments. This paper formalizes macroeconomic stabilization as a Markov Decision Process and investigates whether an RL agent can learn robust policy rules.

2 MDP Formulation

We define the system as

$$s_{t+1} = f(s_t, a_t, \epsilon_t), \quad r_t = R(s_t, a_t), \quad (1)$$

where ϵ_t denotes a stochastic shock. At each time step the agent observes s_t , selects an action a_t , and the environment returns a successor state s_{t+1} together with a scalar reward r_t . Supervised learning provides labelled targets, whereas reinforcement learning specifies only a reward signal; the agent must therefore discover a policy through interaction. The objective is the expected long-run cumulative reward, which makes the framework suitable for sequential decision problems in which present actions influence future opportunities.

2.1 State Space

$$s_t = [\pi_t, u_t, g_t, r_t, d_t, \mathbb{E}_t \pi_{t+1}, \tau_t, G_t]. \quad (2)$$

The state vector summarizes the information available to the agent at time t . It collects the contemporaneous macroeconomic variables — inflation, unemployment, growth, the policy rate, debt, expected inflation, the tax rate, and the level of government spending — and therefore captures both the current macroeconomic situation and the prevailing policy stance.

2.2 Action Space

$$a_t = [\Delta r_t, \Delta G_t, \Delta \tau_t].$$

The action is continuous and represents incremental changes to the three policy instruments. The continuous formulation is consistent with Soft Actor-Critic (SAC), the algorithm employed in later sections, which is designed for continuous control problems.

3 Economic Model

3.1 Aggregate Demand with Lag

$$g_t = \alpha_1 G_{t-1} - \alpha_2 r_{t-1} + \epsilon_t^d. \quad (3)$$

Current GDP growth depends on the previous quarter’s policy stance and a stochastic demand disturbance. Government spending enters with a positive coefficient because public expenditure adds directly to aggregate demand, while the policy rate enters with a negative coefficient because higher borrowing costs suppress consumption and investment. The lag captures the empirical fact that monetary and fiscal interventions transmit gradually through the economy. The shock ϵ_t^d represents unanticipated swings in private demand.

3.2 Expectations-Augmented Phillips Curve

$$\pi_t = \mathbb{E}_t \pi_{t+1} + \beta(g_t - g^*) + \epsilon_t^s. \quad (4)$$

Inflation is driven by expected inflation and by the deviation of growth from its potential rate g^* . When the economy operates above potential, tighter labor markets and stronger pricing power raise inflation; when it operates below potential, these pressures abate. Forward-looking expectations enter the equation because wage and price-setting depend on anticipated future inflation. The supply shock ϵ_t^s accounts for cost-side disturbances such as energy prices or supply-chain disruptions.

3.3 Expectations Formation (Adaptive)

$$\mathbb{E}_t \pi_{t+1} = \rho \pi_t + (1 - \rho) \mathbb{E}_{t-1} \pi_t.$$

Households and firms update their expectations adaptively rather than under full rationality. The parameter ρ governs the speed of adjustment: a larger ρ produces faster updating and stronger persistence of inflation, while a smaller ρ implies sticky beliefs. Persistent expectations feed back into the Phillips curve and propagate inflationary pressure across periods.

3.4 Okun’s Law

$$u_t = u_{t-1} - \gamma(g_t - g^*).$$

Unemployment responds to the deviation of growth from its potential rate. Above-trend growth lowers unemployment, while below-trend growth raises it. The coefficient γ measures the sensitivity of the labor market to growth surprises and underpins the Phillips-curve trade-off: stimulating demand to reduce unemployment also tends to raise inflation.

3.5 Debt Dynamics

$$d_t = d_{t-1} + G_t - \tau_t.$$

The debt-to-GDP ratio evolves according to the simple identity above: deficits accumulate as debt, and surpluses reduce it. Richer specifications would include interest payments and the GDP denominator; the present formulation retains only the qualitative trade-off between short-run fiscal support and long-run debt sustainability.

3.6 Policy Accumulation

$$\begin{aligned} r_t &= r_{t-1} + \Delta r_t, \\ G_t &= G_{t-1} + \Delta G_t, \\ \tau_t &= \tau_{t-1} + \Delta \tau_t. \end{aligned}$$

The agent controls the increments rather than the levels. This convention reflects the incremental nature of real-world policymaking and induces inertia: persistent increments accumulate into a tight or loose stance over time. The accumulation rule renders the environment path-dependent, since each decision shifts the starting point for subsequent ones.

4 Reward Function

In the milestone-one simulator, welfare is scored relative to policy targets:

$$r_t = -w_1(\pi_t - \pi^*)^2 - w_2(u_t - u^*)^2 + w_3(g_t - g^*) - w_4(d_t - d^*). \quad (5)$$

The reward provides the agent’s learning signal. In supervised learning the loss measures the gap between predictions and labels; in reinforcement learning the reward instead encodes a designer-specified notion of good performance. The quadratic penalties on inflation and unemployment imply that large deviations from target are penalized disproportionately, while the linear growth and debt terms reward expansion above potential and discourage debt above the sustainable level. The agent’s optimization target is the expected discounted sum of these rewards over the horizon.

5 SAC Objective

$$J(\pi) = \mathbb{E} \left[\sum_t r_t + \alpha H(\pi(\cdot|s_t)) \right]. \quad (6)$$

The symbol π denotes the policy, that is, the mapping from states to action distributions, and $J(\pi)$ measures the expected long-run performance of π when it interacts with the stochastic environment. The expectation is taken over both policy randomness and environment shocks, so $J(\pi)$ averages performance across many trajectories rather than a single realization.

The first term in $J(\pi)$ is the cumulative reward: a policy that maintains inflation and unemployment near their targets, supports growth, and contains debt achieves a high value. The second term, $H(\pi(\cdot|s_t))$, denotes the entropy of the policy at state s_t . Including the entropy term

encourages continued exploration and prevents premature convergence to a narrow behavior. The temperature coefficient α governs the trade-off between exploitation and exploration.

The objective is rendered tractable by an action-value function. Improving π requires knowing which actions yield high expected return, which is precisely what the critic estimates:

$$Q(s_t, a_t) \approx \mathbb{E} \left[\sum_{k=t}^T \gamma^{k-t} r_k \middle| s_t, a_t \right].$$

Thus $J(\pi)$ defines the global objective, and $Q(s, a)$ provides the local evaluation tool used to update it. The actor moves the policy toward actions with high critic scores while the entropy term preserves exploration.

6 Neural Network Architecture

6.1 Actor (Policy)

- Input: s_t
- Architecture: MLP with 2–3 hidden layers
- Output: Gaussian policy parameters (μ, σ)

The actor is the network that selects actions. It can be viewed as an MLP that maps the state vector to a distribution over actions. SAC parameterizes this distribution as a Gaussian with state-dependent mean μ and scale σ , from which an action is sampled at each step. The stochastic parameterization supports exploration during early training and concentrates around high-value actions as learning progresses.

6.2 Critic (Q-function)

- Input: (s_t, a_t)
- Output: $Q(s_t, a_t)$

The critic is a separate network that evaluates actions. Its target is the action-value function, defined as the expected discounted sum of future rewards obtained by taking action a_t in state s_t and following the current policy thereafter. The discount factor $\gamma \in (0, 1)$ down-weights distant rewards, ensuring convergence of the infinite sum and reflecting greater uncertainty about the long-run future. The critic supplies the long-run evaluation that distinguishes reinforcement learning from one-step regression: actions are judged not only by their immediate consequences but also by the trajectory of future states they induce.

6.3 SAC Architecture Diagram

This figure depicts the SAC training flow for the macroeconomic policy environment. The policy observes the macro state $s_t = [\pi_t, u_t, g_t, r_t, d_t, \mathbb{E}_t \pi_{t+1}, \tau_t, G_t]$ and outputs the three policy increments $(\Delta r_t, \Delta G_t, \Delta \tau_t)$. The environment returns a reward and a successor state, and the resulting transition is stored in the replay buffer. The critic $Q_\theta(s_t, a_t)$ is updated by minimizing the discrepancy with an entropy-adjusted target value built from r_t , s_{t+1} , the target network, and the next-state action sample. The actor draws candidate actions from the current policy, queries the critic, and

SAC relation	Information used	Role in this project
Environment interaction	The policy observes $s_t = [\pi_t, u_t, g_t, r_t, d_t, \mathbb{E}_t \pi_{t+1}, \tau_t, G_t]$ and samples $a_t = (\Delta r_t, \Delta G_t, \Delta \tau_t)$.	The macro environment applies the policy move and returns (r_t, s_{t+1}) . This creates the raw experience used for learning.
Replay buffer	Transitions (s_t, a_t, r_t, s_{t+1}) collected from previous economy rollouts.	SAC samples mini-batches from this memory so the critic and actor learn from many past macroeconomic situations, not only the latest quarter.
Critic update	The critic estimates $Q_\theta(s_t, a_t)$ and compares it with an entropy-adjusted target value built from r_t and s_{t+1} .	This trains the Q-function to judge whether a policy action improves long-run macroeconomic welfare. Practical SAC usually uses two critics to reduce overestimation.
Actor update	The actor samples a candidate action $a'_t \sim \pi_\phi(\cdot s_t)$ and receives the critic score $Q_\theta(s_t, a'_t)$ plus the entropy term.	This shifts the policy toward actions with higher estimated long-run value while preserving exploration through the entropy bonus.

Figure 1: SAC architecture represented as a relation map. This table replaces a dense arrow diagram so that the same dependencies are visible without overlapping labels or crossed feedback arrows.

updates the policy parameters toward higher-value actions while preserving exploration through the entropy term. The figure displays a single critic for clarity; the algorithm in this project follows the standard SAC implementation.

7 Training Algorithm

Soft Actor-Critic for Policy Learning

1. Initialize networks and replay buffer.
2. For each episode:
 - (a) Reset the environment.
 - (b) For $t = 1$ to T :
 - i. Sample action $a_t \sim \pi(s_t)$.
 - ii. Observe s_{t+1} and r_t .
 - iii. Store the transition.
 - iv. Update the critic via Bellman loss.
 - v. Update the actor via policy gradient.

The training loop integrates the components described above. Each interaction with the environment produces a transition tuple (s_t, a_t, r_t, s_{t+1}) , which is appended to the replay buffer. SAC samples mini-batches from the buffer to update both networks via stochastic gradient descent: the critic minimizes a Bellman-style temporal-difference loss against a bootstrapped target, and the

actor is updated to favor actions that the critic scores highly while preserving entropy. Repeated interleaving of these updates and environment interactions drives the joint convergence of value estimates and policy.

8 Experimental Design

8.1 Baselines

- Taylor rule
- Random policy

8.2 Shock Scenarios

- Demand shocks (ϵ^d)
- Supply shocks (ϵ^s)
- Structural breaks

8.3 Metrics

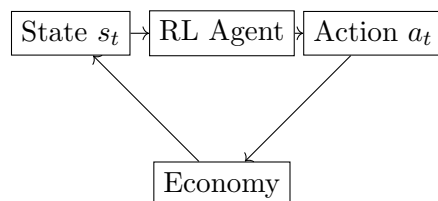
- Variance of inflation
- Variance of unemployment
- Mean GDP growth
- Debt trajectory
- Cumulative welfare

9 Ablation Studies

We analyze:

- Effect of removing expectations
- Effect of removing lag structure
- Sensitivity to reward weights

10 System Diagram



11 Discussion

This framework highlights the role of learning-based control in macroeconomics. Challenges include model misspecification, reward alignment, and generalization under structural change.

12 From Macroeconomic Control to a General Failure Question

The macroeconomic simulator constitutes one instance of a broader class of controlled Markov systems. In the general setting considered below, the state is a vector $s_t \in \mathbb{R}^n$, the action is produced by a policy π , and the system evolves according to

$$s_{t+1} = f(s_t, \pi(s_t)).$$

The control objective is a prescribed target s^* , with performance measured by the quadratic loss

$$L(s_t, s^*) = \|s_t - s^*\|_2^2. \quad (7)$$

The central question is therefore not only whether reinforcement learning can reduce this loss in practice, but under what structural conditions *any* admissible policy can drive the trajectory to the target. Failure may originate in the system itself, independently of the optimization procedure.

12.1 Equilibrium Feasibility

Stabilization analysis presupposes that the target can be maintained once attained. Otherwise the loss cannot converge to zero, since s^* is not a fixed point of the controlled dynamics. Let u^* denote the action applied at the target. A necessary feasibility condition is

$$s^* = f(s^*, u^*), \quad u^* \in \mathcal{U},$$

where $\mathcal{U}$ denotes the admissible action set. This condition separates two distinct sources of failure: optimization, which concerns whether an algorithm can discover a good policy; and feasibility, which concerns whether the target is compatible with the constraints of the system. When no admissible u^* satisfies the fixed-point equation, even an ideal optimizer cannot drive $L(s_t, s^*)$ to zero.

12.2 Lyapunov Analysis

Reward trajectories alone do not establish stability. A reinforcement learning agent may obtain temporarily favorable rewards while the underlying state remains fragile, oscillatory, or only conditionally stable. The Lyapunov framework addresses this gap by providing a direct mathematical test for whether deviations from the target contract over time.

For a fixed policy, the controlled system reduces to the closed-loop map

$$s_{t+1} = g(s_t) := f(s_t, \pi(s_t)).$$

The natural Lyapunov candidate is the loss itself,

$$V(s_t) = L(s_t, s^*) = \|s_t - s^*\|_2^2,$$

which links the optimization and stability objectives. Since V is positive definite about s^* , attention turns to its one-step difference

$$\Delta V(s_t) = V(s_{t+1}) - V(s_t) = \|g(s_t) - s^*\|_2^2 - \|s_t - s^*\|_2^2.$$

The sign of ΔV indicates the direction of the error energy along trajectories. If $\Delta V(s_t) < 0$ for every s_t in a neighborhood of s^* except the target itself, the closed-loop system is locally asymptotically stable in that neighborhood. Directions in which $\Delta V \geq 0$ correspond to a failure of the policy to contract deviations.

12.3 Local Linearization

For high-dimensional nonlinear systems the exact form of ΔV is rarely tractable. A first-order expansion around the target yields a linear approximation in which the source of failure becomes algebraically explicit.

Define the local deviations

$$\Delta s_t = s_t - s^*, \quad \Delta u_t = u_t - u^*.$$

Assuming differentiability of f at (s^*, u^*) , a first-order Taylor expansion produces the linearized dynamics

$$\Delta s_{t+1} \approx A \Delta s_t + B \Delta u_t, \tag{8}$$

with Jacobian matrices

$$A = \left. \frac{\partial f}{\partial s} \right|_{(s^*, u^*)}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{(s^*, u^*)}.$$

The matrix A encodes the endogenous propagation of deviations under a fixed action, while B measures the influence of the control channel on the state. Intuitively, A captures persistence and internal instability, while B captures control authority.

12.4 The Closed-Loop Matrix $G = A + BK$

The linearized dynamics still depend on the action increment. Approximating the policy locally as a linear feedback,

$$\Delta u_t \approx K \Delta s_t, \quad K = \left. \frac{\partial \pi}{\partial s} \right|_{s^*},$$

substitutes the policy into the dynamics and yields the closed-loop recursion

$$\Delta s_{t+1} \approx G \Delta s_t, \quad G = A + BK. \tag{9}$$

The matrix G aggregates the dynamics and the policy into a single object whose spectral properties determine local stability.

For the quadratic candidate $V(\Delta s_t) = \|\Delta s_t\|_2^2$, the Lyapunov difference admits an explicit form. Substituting the linearized closed-loop recursion $\Delta s_{t+1} \approx G \Delta s_t$ into the candidate gives

$$V(\Delta s_{t+1}) = \Delta s_{t+1}^\top \Delta s_{t+1} \approx (G \Delta s_t)^\top (G \Delta s_t) = \Delta s_t^\top G^\top G \Delta s_t,$$

so that

$$\Delta V(\Delta s_t) = V(\Delta s_{t+1}) - V(\Delta s_t) \approx \Delta s_t^\top G^\top G \Delta s_t - \Delta s_t^\top \Delta s_t = \Delta s_t^\top (G^\top G - I) \Delta s_t. \quad (10)$$

This identity converts the stabilization requirement from a property of the nonlinear map into an algebraic property of the matrix $G^\top G - I$. The quadratic form is strictly negative for all nonzero Δs_t if and only if $G^\top G - I \prec 0$, equivalently if all singular values of G are strictly less than one. Geometrically, ΔV is negative when G contracts the Euclidean ball at s^* in every direction. Stabilization with the unweighted Euclidean candidate therefore demands not only $\rho(G) < 1$ but the stronger contraction condition $\|G\|_2 < 1$. More generally, local asymptotic stability is characterized by

$$\rho(G) < 1, \quad (11)$$

where $\rho(G)$ denotes the spectral radius. Equivalently, there exists a positive definite matrix P such that $G^\top P G - P \prec 0$. The general weighted form is preferred when state variables differ in scale or interpretation, as is typical in macroeconomic systems.

12.5 Controllability

Local stability through $\rho(G) < 1$ requires that an appropriate feedback gain K exists. Controllability formalizes whether the inputs can influence all relevant state directions of the linearized system.

Starting from the linearized dynamics $\Delta s_{t+1} = A\Delta s_t + B\Delta u_t$, repeated substitution propagates initial conditions and inputs forward in time. Substituting the recursion into itself once gives

$$\Delta s_2 = A(A\Delta s_0 + B\Delta u_0) + B\Delta u_1 = A^2\Delta s_0 + AB\Delta u_0 + B\Delta u_1,$$

and iterating once more yields

$$\Delta s_3 = A^3\Delta s_0 + A^2B\Delta u_0 + AB\Delta u_1 + B\Delta u_2. \quad (12)$$

The structure visible at step three generalizes by induction. Each application of the recursion advances the homogeneous term by one power of A and prepends an additional input term $B\Delta u_k$ premultiplied by the appropriate power of A that accounts for the remaining transitions. Continuing to step n therefore yields

$$\Delta s_n = A^n\Delta s_0 + A^{n-1}B\Delta u_0 + A^{n-2}B\Delta u_1 + \cdots + B\Delta u_{n-1}.$$

This expansion exposes how present and past actions propagate through the coupled dynamics. Rearranging produces

$$\Delta s_n - A^n\Delta s_0 = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} \begin{bmatrix} \Delta u_{n-1} \\ \Delta u_{n-2} \\ \vdots \\ \Delta u_0 \end{bmatrix}.$$

The block matrix

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} \quad (13)$$

is the Kalman controllability matrix. Its column space describes all state directions reachable through admissible inputs across multiple time steps.

When

$$\text{rank}(\mathcal{C}) < n, \quad (14)$$

some state directions are unreachable under the linearized dynamics, and no local feedback policy can fully counteract disturbances along those directions. Even at full rank, control authority may still be insufficient when the entries of B are small or when the constraints in $\mathcal{U}$ bind tightly. The system is then controllable in principle but too weakly actuated in practice to overcome shocks, nonlinear drift, or saturation. This formalizes the intuition that a volatile system can fail when policy influence is small relative to endogenous motion.

12.6 A Worked Linear Macroeconomic Example

The previous subsections introduced the general tools. A concrete example is needed because it shows how an apparently reasonable economic model can contain a structural failure mode before any RL algorithm is trained. The simplified macroeconomic system in this paper is already close to linear, so it can be rewritten as an explicit state-space model and analyzed directly.

For this analysis, stochastic shocks are removed and all variables are expressed as deviations from a steady state. The transformation does not alter the economic content; it is a coordinate change that renders stability and controllability transparent. Let the steady state be

$$(g^*, \pi^*, u^*, d^*, r^*, G^*, \tau^*, e^*),$$

where $e_t = \mathbb{E}_t \pi_{t+1}$. A feasible steady state must satisfy

$$\begin{aligned} g^* &= \alpha_1 G^* - \alpha_2 r^*, \\ \pi^* &= e^*, \\ G^* &= \tau^*. \end{aligned}$$

The first condition requires that trend growth be consistent with the steady policy stance. The second equates inflation with its expectation, so that no anticipated correction remains. The third holds debt constant only when government spending equals tax revenue in this simplified accounting.

Define deviation variables by adding a tilde to each variable:

$$\begin{aligned} \tilde{g}_t &= g_t - g^*, & \tilde{\pi}_t &= \pi_t - \pi^*, & \tilde{u}_t &= u_t - u^*, \\ \tilde{d}_t &= d_t - d^*, & \tilde{r}_t &= r_t - r^*, & \tilde{G}_t &= G_t - G^*, \\ \tilde{\tau}_t &= \tau_t - \tau^*, & \tilde{e}_t &= e_t - e^*. \end{aligned}$$

In these coordinates, the deterministic dynamics read

$$\begin{aligned} \tilde{g}_t &= \alpha_1 \tilde{G}_{t-1} - \alpha_2 \tilde{r}_{t-1}, \\ \tilde{\pi}_t &= \tilde{e}_t + \beta \tilde{g}_t, \\ \tilde{e}_t &= \rho \tilde{\pi}_t + (1 - \rho) \tilde{e}_{t-1}, \\ \tilde{u}_t &= \tilde{u}_{t-1} - \gamma \tilde{g}_t, \\ \tilde{d}_t &= \tilde{d}_{t-1} + \tilde{G}_t - \tilde{\tau}_t. \end{aligned}$$

The expectation and inflation equations are simultaneous; eliminating $\tilde{\pi}_t$ before assembling the system matrix yields a cleaner representation. Substituting $\tilde{e}_t = \rho\tilde{\pi}_t + (1 - \rho)\tilde{e}_{t-1}$ into $\tilde{\pi}_t = \tilde{e}_t + \beta\tilde{g}_t$ produces

$$\tilde{\pi}_t = \rho\tilde{\pi}_t + (1 - \rho)\tilde{e}_{t-1} + \beta\tilde{g}_t,$$

hence

$$\tilde{\pi}_t = \tilde{e}_{t-1} + \frac{\beta}{1 - \rho}\tilde{g}_t.$$

Defining

$$\kappa = \frac{\beta}{1 - \rho}, \quad \eta = \frac{\rho\beta}{1 - \rho},$$

the expectation equation reduces to

$$\tilde{e}_t = \tilde{e}_{t-1} + \eta\tilde{g}_t.$$

This reduction shows that the persistent memory of the system resides in expected inflation rather than in inflation itself, and that expectations are driven by the same output-gap channel that drives unemployment.

Define the state vector and policy action by

$$x_t = \begin{bmatrix} \tilde{u}_{t-1} \\ \tilde{e}_{t-1} \\ \tilde{d}_{t-1} \\ \tilde{r}_{t-1} \\ \tilde{G}_{t-1} \\ \tilde{\tau}_{t-1} \end{bmatrix}, \quad a_t = \begin{bmatrix} \Delta r_t \\ \Delta G_t \\ \Delta \tau_t \end{bmatrix}.$$

The output gap and inflation deviation are linear in the state:

$$\begin{aligned} \tilde{g}_t &= [0 \quad 0 \quad 0 \quad -\alpha_2 \quad \alpha_1 \quad 0] x_t, \\ \tilde{\pi}_t &= [0 \quad 1 \quad 0 \quad -\kappa\alpha_2 \quad \kappa\alpha_1 \quad 0] x_t. \end{aligned}$$

The full linear state-space system is

$$x_{t+1} = Ax_t + Ba_t, \tag{15}$$

where

$$A = \begin{bmatrix} 1 & 0 & 0 & \gamma\alpha_2 & -\gamma\alpha_1 & 0 \\ 0 & 1 & 0 & -\eta\alpha_2 & \eta\alpha_1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The structure of these matrices reflects the economic ordering of the model: rate, spending, and tax adjustments first move the corresponding policy states, which subsequently propagate to output and from there to unemployment, expectations, and debt.

12.7 Kalman Interpretation of the Linear Example

The Kalman analysis asks whether policy actions can influence every independent state direction. For the six-dimensional system the controllability matrix is

$$\mathcal{C} = [B \quad AB \quad A^2B \quad A^3B \quad A^4B \quad A^5B].$$

For generic nonzero parameters of the simplified model, this matrix has rank five rather than six:

$$\text{rank}(\mathcal{C}) = 5 < 6. \quad (16)$$

The rank deficiency admits a closed-form explanation. The unemployment and expectation equations

$$\begin{aligned} \tilde{u}_t &= \tilde{u}_{t-1} - \gamma \tilde{g}_t, \\ \tilde{e}_t &= \tilde{e}_{t-1} + \eta \tilde{g}_t, \end{aligned}$$

combined as η times the first plus γ times the second, yield

$$\eta \tilde{u}_t + \gamma \tilde{e}_t = \eta \tilde{u}_{t-1} + \gamma \tilde{e}_{t-1},$$

so the quantity

$$q_t = \eta \tilde{u}_t + \gamma \tilde{e}_t \quad (17)$$

is an invariant of the controlled dynamics. No admissible action sequence can change its value. In vector form, with

$$w = [\eta \quad \gamma \quad 0 \quad 0 \quad 0 \quad 0]^\top,$$

the invariant satisfies $w^\top x_t = q_t$, and the system matrices obey

$$w^\top A = w^\top, \quad w^\top B = 0. \quad (18)$$

This is precisely the structural obstruction detected by Kalman controllability. Because one linear combination of the state is preserved under every admissible action, the reachable set from initial condition x_0 lies on the hyperplane $w^\top x = w^\top x_0$. The origin is therefore attainable only when $w^\top x_0 = 0$. If

$$\eta \tilde{u}_0 + \gamma \tilde{e}_0 \neq 0,$$

no policy, learned or hand-designed, can drive all six deviations to zero.

The economic interpretation is direct. Unemployment and expected inflation respond to the same output-gap channel, while policy can affect output only through rates and spending. The model contains three policy instruments but only one effective channel into the (u, e) subspace, leaving a one-dimensional uncontrollable direction.

12.8 Lyapunov Interpretation of the Linear Example

The Lyapunov perspective produces the same conclusion from the stability side. For a local linear policy $a_t = Kx_t$, the closed-loop system is

$$x_{t+1} = (A + BK)x_t = Gx_t. \quad (19)$$

Asymptotic stability at the origin in the full six-dimensional state space requires $\rho(G) < 1$, equivalently the existence of a positive definite matrix P with $G^\top PG - P \prec 0$.

The invariant identified above persists under every feedback gain K . Since $w^\top B = 0$,

$$w^\top G = w^\top (A + BK) = w^\top A + w^\top BK = w^\top,$$

so 1 is a left eigenvalue of G for every linear feedback law. Consequently

$$\rho(G) \geq 1 \quad (20)$$

universally. No linear policy can render the entire six-dimensional deviation vector asymptotically stable at the origin. A Lyapunov function may decrease on the five-dimensional controllable subspace, but no quadratic Lyapunov function can be strictly decreasing in all nonzero directions of the full state space.

The two perspectives connect in a single chain:

$$\begin{aligned} \text{structural invariant} &\implies \text{rank}(\mathcal{C}) < 6 \implies \text{immovable closed-loop mode} \\ &\implies \rho(G) \geq 1 \implies \text{no full-state Lyapunov proof of asymptotic stability.} \end{aligned}$$

The example provides a useful counterexample for the research question of this paper. Even a linear, low-dimensional, economically interpretable model need not be stabilizable to an arbitrary target from an arbitrary initial condition. Failure of an RL policy in such a setting is not attributable to insufficient training or limited network capacity; it reflects the geometry of the controlled Markov process itself.

12.9 Constraint Sets and Viability

In constrained systems, asymptotic convergence does not exhaust the relevant criteria. A policy may drive the state toward the target in the long run while violating hard limits along the way. Viability theory addresses this gap by distinguishing asymptotic desirability from trajectory-wise admissibility.

Let $\mathcal{X} \subseteq \mathbb{R}^n$ denote the safe set, namely the set of states consistent with physical, institutional, or regulatory constraints. The viability kernel is

$$\text{Viab}_f(\mathcal{X}) = \{s_0 \in \mathcal{X} \mid \exists \pi \text{ such that } s_t \in \mathcal{X} \text{ for all } t \geq 0\}. \quad (21)$$

The kernel identifies the initial states that are recoverable under the prescribed control limits. When $s_0 \notin \text{Viab}_f(\mathcal{X})$, failure is unavoidable: every admissible policy forces the trajectory out of $\mathcal{X}$ before stabilization can occur. In the macroeconomic context, this corresponds to initial conditions so extreme that no bounded sequence of policy actions can restore the system without first breaching debt, inflation, or unemployment limits.

12.10 Chain of Failure

The preceding concepts form a single logical chain. Failure of RL-based stabilization does not originate solely in the optimization step; it may arise upstream, at the level of feasibility, control authority, or admissible trajectories.

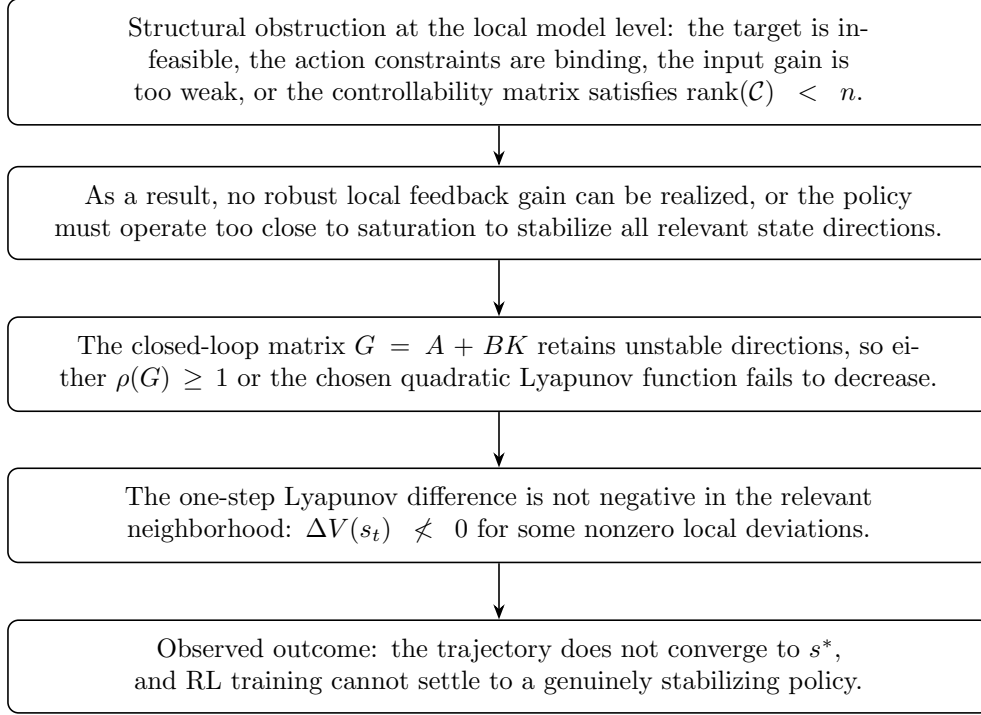


Figure 2: Logical chain linking structural properties of the controlled system to failure of RL-based stabilization.

12.11 Connection Back to the Present Project

The macroeconomic environment developed in this project provides a useful testbed: the state variables carry direct economic interpretation and the action channels are economically meaningful. The reward already resembles a quadratic stabilization objective, which admits a natural Lyapunov-style reading. The preceding analysis nonetheless clarifies an important caveat: favorable RL performance in the simulator does not establish stabilizability of arbitrary coupled Markov systems.

The broader research program is therefore to identify when stabilization fails for structural reasons. Four failure modes deserve emphasis. First, the target s^* may be infeasible under the admissible action set. Second, the local control authority encoded by B may be too weak relative to the intrinsic dynamics encoded by A . Third, the initial state s_0 may lie outside the viability kernel of the constrained system. Fourth, strong nonlinearities may shrink the domain in which the local approximation is valid, so that a policy that stabilizes near s^* still fails from more distant initial conditions. These conditions delineate the boundary between a trainable controller and a genuinely stabilizing one.

13 Selected Theoretical Literature

The present project sits at the intersection of reinforcement learning, nonlinear control, and constrained dynamical systems. Several references are especially relevant to the failure question posed above.

- Tsitsiklis and Van Roy, “Analysis of Temporal-Difference Learning with Function Approximation” (1996), for early mathematical results showing that value-function learning with

function approximation can be unstable.

- Dai et al., “SBEED: Convergent Reinforcement Learning with Nonlinear Function Approximation” (ICML 2018), for a route toward convergence guarantees under nonlinear approximation.
- Zhang et al., “Provably Convergent Two-Timescale Off-Policy Actor-Critic with Function Approximation” (ICML 2020), for actor-critic convergence theory closer to modern RL practice.
- Boffi et al., “Learning Stability Certificates from Data” (CoRL 2021), for learning Lyapunov-style certificates directly from trajectories.
- Wan, Naik, and Sutton, “Learning and Planning in Average-Reward Markov Decision Processes” (ICML 2021), for a formulation that is often more appropriate than discounted return in continuing control systems.
- Moerland et al., “Model-based Reinforcement Learning: A Survey” (2022), for a broad overview of approaches that may be better suited to structured dynamical systems than purely model-free methods.

14 Conclusion

We demonstrate a reinforcement learning approach to macroeconomic policy design. Future work includes multi-agent economies and real-world calibration.